Q4. Open ports and running services

Original question: Scan an assigned lab machine to identify open ports and running services. Interpret the scan output and identify potential security concerns.

Aim

Scan one assigned lab machine to identify exposed TCP ports and likely services, then interpret potential concerns.

Requirements

Nmap, the assigned machine's IP, permitted port scope and connectivity to the lab network. Use one host only. Reference: [R3].

Procedure

Step 1 - Confirm scope and route

```
mkdir -p "$HOME/EHF_Exam/Q4"
cd "$HOME/EHF_Exam/Q4"
TARGET='<LAB_TARGET_IP>'
date -Is | tee collected_at.txt
nmap --version
ip route get "$TARGET"
```

`ip route get` shows the chosen route, interface and source address. It does not prove that the target is reachable. Confirm the VM network mode and IP with the examiner.

Step 2 - Scan the default TCP port set

```
nmap -n -sT --reason -oN q4_ports.txt "$TARGET"
cat q4_ports.txt
```

`-sT` uses TCP connections and normally works without root. `-n` avoids reverse-DNS lookups. Without a port option, Nmap normally scans its 1,000 most common TCP ports, not all 65,535. `--reason` explains state decisions; `-oN` saves readable output. The SERVICE column without -sV is usually a port-based label, not verified software identification.

Step 3 - Confirm services on the observed open ports

```
PORTS='<OBSERVED_OPEN_TCP_PORTS>'
nmap -n -sT -sV --version-light -p "$PORTS" \
  -oN q4_services.txt "$TARGET"
```

Replace PORTS with comma-separated numbers from Step 2, such as 22,80 only if those were observed. `-sV` probes service responses; `--version-light` uses fewer probes. If host discovery is blocked but the assigned host is known to be up, repeat the approved scan with `-Pn`; this skips discovery and can take longer. Do not expand the host or port scope without lab authorization.

Expected output - illustrative

```
PORT    STATE     SERVICE
22/tcp  open      ssh
80/tcp  open      http
443/tcp filtered  https
```

This is a shortened synthetic port table; it is not a scan of your target. An open port accepted a connection; closed generally means no listener responded at that port; filtered means filtering or missing responses prevented a clear decision.

Interpretation, observation and result

Record port, state, reason, service evidence and concern. An exposed administration service may need access restrictions; HTTP may transmit unencrypted information; neither finding proves exploitation. Only call a service unnecessary if the machine's role supports that conclusion. Record ports not scanned and distinguish filtering from closure. The practical is complete when the saved scan is interpreted and observed service labels are separated from confirmed responses.

Common errors

- Host seems down: verify IP/VM networking; use -Pn only for the known assigned host.
- Empty results: examine the full output and port scope; no open ports in this scan is a valid result.
- Permission complaint: use the explicit -sT example rather than a privileged raw-packet scan.

Viva questions

- What is a port? A transport-layer endpoint number used with an address and protocol.
- Does open mean vulnerable? No; it means a listener is reachable.
- Why add -sV? To obtain service-response evidence instead of relying only on the port number.
- What does filtered mean? Nmap cannot determine open/closed because probes or responses are blocked or absent.